

Smart Metro Ticket Booking System

Requirement Analysis Phase

Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	4 Marks

2. Data Flow Diagram & User Stories

This document presents the system's data flow at Level 0 (context) and Level 1 (process decomposition), followed by user stories capturing functional needs from the perspective of each identified persona: Passengers, Station Managers, the Metro Operations Team, and IT Admins.

2.1 Data Flow Diagram — Level 0 (Context)

Data Flow Diagram — Level 0 (Context Diagram)

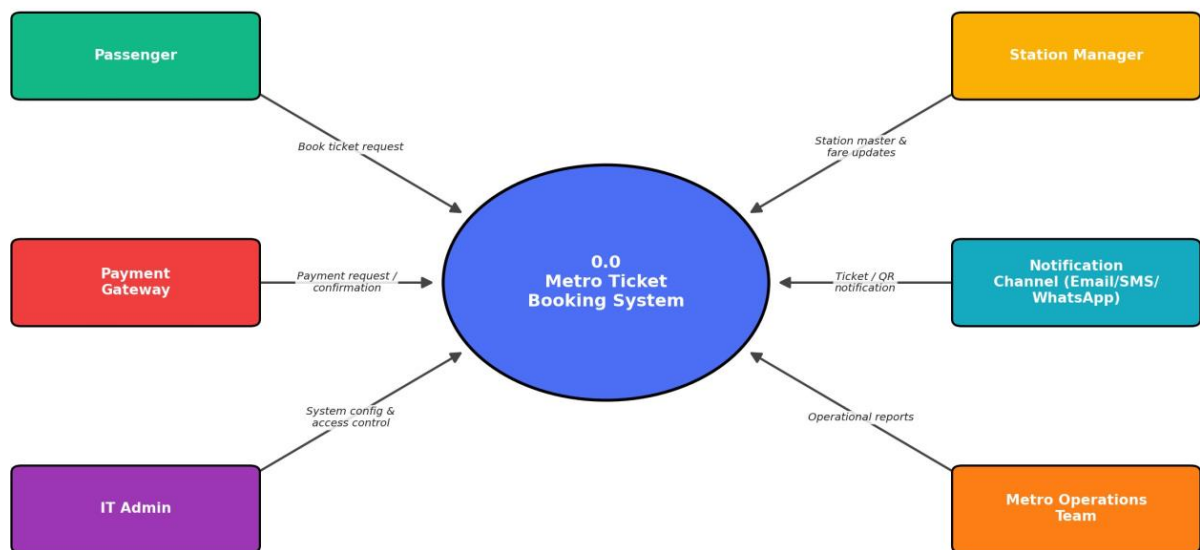


Figure 2.1 — External entities interacting with the Metro Ticket Booking System

The context diagram identifies four human actors and two external systems that exchange data with the core booking platform: Passengers initiate booking requests; Station Managers maintain station and fare master data; the Metro Operations Team consumes operational reports; IT Admins manage configuration and access; the Payment Gateway processes transactions; and the Notification Channel delivers tickets and alerts.

2.2 Data Flow Diagram — Level 1 (Process Decomposition)

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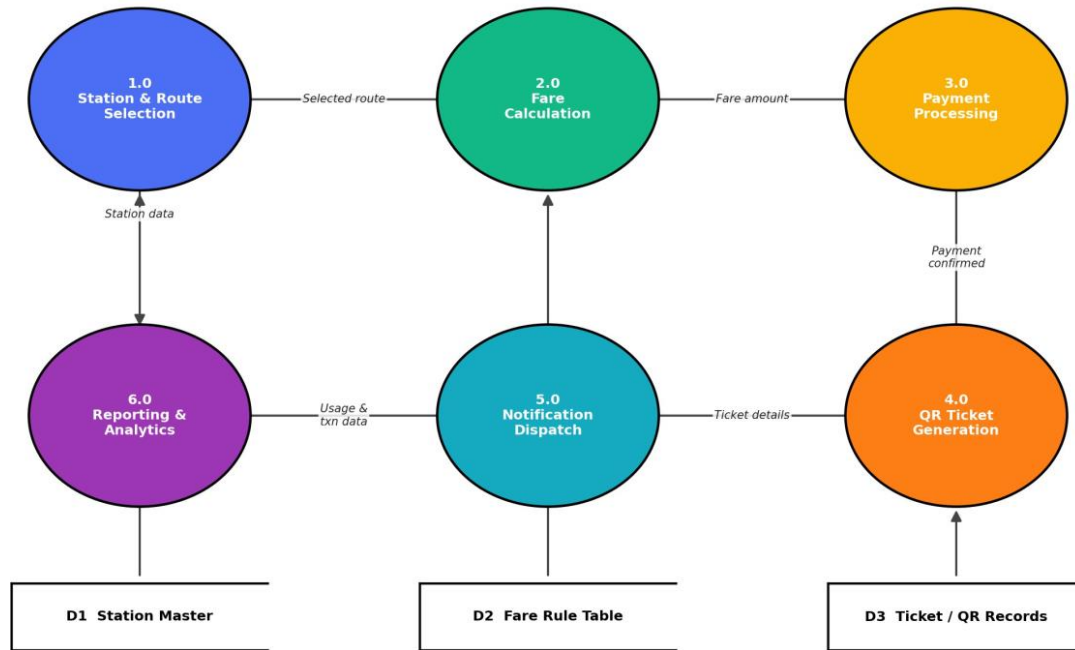


Figure 2.2 — Decomposition of the booking system into six core processes

The system decomposes into six processes: Station & Route Selection, Fare Calculation, Payment Processing, QR Ticket Generation, Notification Dispatch, and Reporting & Analytics. These map directly to Flow Designer sub-flows and Script Includes, reading from and writing to three core data stores — Station Master, Fare Rule Table, and Ticket/QR Records.

2.3 User Stories

User stories are grouped by persona and written in the standard "As a <role>, I want <capability>, so that <benefit>" format, each with supporting acceptance criteria.

Passenger (End User)

ID	User Story	Acceptance Criteria
US-01	As a passenger, I want to select my source and destination stations from a searchable catalog form, so that I can book a ticket without visiting a counter.	Catalog form loads station list from Station Master; invalid source/destination pairs are blocked; search-as-you-type is supported.
US-02	As a passenger, I want to see the calculated fare instantly after entering my travel details, so that I know the exact amount before paying.	Fare updates in real time when station, passenger type, or ticket count changes; discount for Student/Senior Citizen is auto-applied.
US-03	As a passenger, I want to pay digitally using UPI, card, or wallet, so that I don't need exact cash.	Payment sub-flow supports at least three digital modes; failed payments show a clear retry option.
US-04	As a passenger, I want to receive my ticket as a QR code immediately after payment, so that I can travel without printed tickets.	QR is generated within seconds of payment confirmation and delivered via app notification and email/WhatsApp.

ID	User Story	Acceptance Criteria
US-05	As a passenger, I want to view my past bookings and download tickets again, so that I have a record for reimbursement or reuse.	Booking history is available in the Service Portal with filter by date and route.

Station Manager

ID	User Story	Acceptance Criteria
US-06	As a station manager, I want to maintain and update station details and fare zones, so that booking data stays accurate.	Station Master records are editable through a controlled catalog/table with approval workflow for changes.
US-07	As a station manager, I want to validate QR tickets at the gate in real time, so that only paid passengers can enter.	QR scan returns valid/invalid/used status within 2 seconds; duplicate scans are flagged.
US-08	As a station manager, I want a dashboard of today's ticket volume at my station, so that I can plan staffing and manage crowding.	Dashboard refreshes in near real time and is filterable by hour.

Metro Operations Team

ID	User Story	Acceptance Criteria
US-09	As a metro operations manager, I want consolidated analytics on peak travel times and popular routes, so that I can plan schedules and resources.	Reports/Performance Analytics dashboards aggregate booking data by route, time band, and passenger type.
US-10	As a metro operations manager, I want visibility into revenue collected via digital payments versus cash, so that I can track digital adoption progress.	Revenue split report is available monthly and exportable.
US-11	As a metro operations manager, I want automated notifications for failed or disputed transactions, so that I can resolve issues proactively.	Flow Designer triggers an incident/case record for any payment exception.

IT Admin

ID	User Story	Acceptance Criteria
US-12	As an IT admin, I want to configure catalog variables and fare rules without code changes, so that business changes can be made quickly.	Fare Rule Table and catalog variables are admin-configurable via ServiceNow UI.

ID	User Story	Acceptance Criteria
US-13	As an IT admin, I want role-based access control across passenger, station manager, and operations views, so that data is only visible to authorized users.	ACLs restrict table and UI Page access strictly by assigned role.
US-14	As an IT admin, I want monitoring and logs for the QR generation and payment integration, so that I can quickly troubleshoot failures.	Integration Hub spokes log request/response with correlation IDs for traceability.